COTS Fine-Scale SDM: Comprehensive Validation Report

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Overview

This report validates the XGBoost COTS Fine-Scale Species Distribution Model using historical Manta Tow and Cull data. We compare both the **2025 Year-Specific** and **Year-Agnostic** models.

The validation involves: 1. **Grid-Based Historical Validation (100m)**: General statistics (ROC, AUC, Confusion Matrix) evaluating the model against historical COTS/scar presence. 2. **Tow-Path Simulation**: Operational simulation showing which manta tows would be suggested based on the model. 3. **Cull Dive Financial Savings**: Quantification of dive hours and dollars saved by avoiding ineffective culls (0.00 or <0.02 CPUE). 4. **Reef Maps**: Visual mapping of predictions vs. operations.

```r
library(terra)
library(sf)
library(dplyr)
library(readr)
library(readxl)
library(ggplot2)
library(tidyr)
library(knitr)
library(stringr)
library(pROC)

# --- Configuration ---
predict_agnostic_path <- "outputs/COTS_prob_0.02_cpue_agnostic.tif"
predict_year2025_path <- "outputs/COTS_prob_0.02_cpue_year2025_clean.tif"  # native 30m raste
predict_year2025_rg_path <- "outputs/COTS_prob_0.02_cpue_year2025_withRG.tif"
predict_maxent_path    <- "outputs/COTS_maxent_suitability.tif"
manta_csv <- "data/COTS Program  Manta Tow Data-2026-02-04.csv"
cull_xlsx <- "data/250929_COTS-Manta-Cull-RHIS-Data-Matthews-and-Schlawinsky.xlsx"
cull_polys_gpkg <- "data/cull_site_polygons.gpkg"

out_dir <- "validation_outputs"
dir.create(out_dir, showWarnings = FALSE, recursive = TRUE)

# Selected Reefs
reef_select <- c(
  "Tongue Reef (16-026)",
  "Batt Reef (16-029)",
  "Arlington Reef (16-064)",
  "(Big) Broadhurst Reef (No 1) (18-100a)",
  "Britomart Reef (18-024)",
  "Sudbury Reef (No 1) (17-001a)",
  "Trunk Reef (18-027)",
  "Bramble Reef (18-029)",
  "Davies Reef (18-096)",
  "Heron Reef (23-052a)"
)
```

1. Load and Prepare Data

```
# Load Model Predictions
prob_agnostic <- if(file.exists(predict_agnostic_path)) terra::rast(predict_agnostic_path) el
prob_year2025 <- if(file.exists(predict_year2025_path)) terra::rast(predict_year2025_path) el
prob_2025_rg  <- if(file.exists(predict_year2025_rg_path)) terra::rast(predict_year2025_rg_pa
prob_maxent   <- if(file.exists(predict_maxent_path))    terra::rast(predict_maxent_path) els

if(is.null(prob_year2025) || is.null(prob_agnostic)) stop("Missing core prediction rasters.")

# Load Manta Tow Data
manta_raw <- read_csv(manta_csv, show_col_types = FALSE) %>%
  filter(
    !is.na(StartLongitude), !is.na(StartLatitude), !is.na(EndLongitude), !is.na(EndLatitude)
    StartLongitude != 0, StartLatitude != 0, EndLongitude != 0, EndLatitude != 0
  ) %>%
  mutate(
    cots_observed = CrownOfThornsStarfishCount > 0,
    scars_observed = !is.na(ScarsCount) & ScarsCount > 0,
    presence_any = cots_observed | scars_observed,
    ReefName = as.character(ReefName)
  )

# Convert Manta Tows to spatial lines
make_tow_lines <- function(df) {
  lines <- lapply(seq_len(nrow(df)), function(i) {
    p1 <- c(df$StartLongitude[i], df$StartLatitude[i])
    p2 <- c(df$EndLongitude[i], df$EndLatitude[i])
    st_linestring(rbind(p1, p2))
  })
  st_sf(df, geometry = st_sfc(lines, crs = 4326))
}

manta_sf <- make_tow_lines(manta_raw)
manta_sf <- st_transform(manta_sf, crs = crs(prob_year2025))
manta_selected <- manta_sf %>% filter(ReefName %in% reef_select)

# Load Cull Data
cull_raw <- read_excel(cull_xlsx, sheet = 4) %>%
  filter(!is.na(Longitude), !is.na(Latitude), !is.na(Bottomtime), Bottomtime > 0, Longitude
  filter(as.Date(SurveyDate) > as.Date("2018-11-30")) %>%
  mutate(
```

```r
    Total = Cohort1 + Cohort2 + Cohort3 + Cohort4,
    CPUE = Total / Bottomtime,
    is_ghost = (Total == 0),
    is_non_problematic = (CPUE < 0.02),
    ReefName = as.character(ReefName)
  )

# Top 30 reefs with most culling hours in 2025
reef_select_eval <- cull_raw %>%
  filter(as.Date(SurveyDate) >= as.Date("2025-01-01")) %>%
  group_by(ReefName) %>%
  summarise(Total_Hours = sum(Bottomtime, na.rm=TRUE) / 60) %>%
  arrange(desc(Total_Hours)) %>%
  slice_head(n = 30) %>%
  pull(ReefName)

# Manta tows for the selected 10 mapping reefs AND the 30 evaluated reefs
manta_selected <- manta_sf %>% filter(ReefName %in% reef_select)
manta_eval_all <- manta_sf %>% filter(ReefName %in% reef_select_eval)

# Cull dives for the selected 10 mapping reefs
cull_selected <- cull_raw %>% filter(ReefName %in% reef_select)
cull_sf <- st_as_sf(cull_selected, coords = c("Longitude", "Latitude"), crs = 4326) %>%
  st_transform(crs = crs(prob_year2025))

# Cull dives for the 30 evaluated reefs (point geometry)
cull_sf_eval_all <- st_as_sf(cull_raw %>% filter(ReefName %in% reef_select_eval), coords = c
  st_transform(crs = crs(prob_year2025))

# Load cull site polygons and join to cull data
cull_site_polys <- st_read(cull_polys_gpkg, quiet = TRUE) %>%
  st_transform(crs = crs(prob_year2025))

cat("Loaded", nrow(cull_site_polys), "cull site polygons from gpkg\n")
```

```
Loaded 15884 cull site polygons from gpkg
```

```r
cat("Matching to", n_distinct(cull_raw$CullSiteName), "unique cull site names...\n")
```

```
Matching to 3729 unique cull site names...
```

```r
cat("Match rate:", round(mean(cull_raw$CullSiteName %in% cull_site_polys$CullSiteName) * 100
```

```
Match rate: 100 %
```

```r
# Calculate Default Threshold: Median of 2025 raster cropped to selected reefs
ext_val <- terra::ext(manta_selected) + 2000
crop_2025 <- terra::crop(prob_year2025, ext_val)
```

```
|---------|---------|---------|---------|
=========================================
```

```r
median_thresh_2025 <- as.numeric(terra::global(crop_2025, fun=function(x) median(x, na.rm=TRU

crop_agnostic <- terra::crop(prob_agnostic, ext_val)
```

```
|---------|---------|---------|---------|
=========================================
```

```r
median_thresh_agnostic <- as.numeric(terra::global(crop_agnostic, fun=function(x) median(x, n
crop_maxent <- if(!is.null(prob_maxent)) terra::crop(prob_maxent, ext_val) else NULL
```

```
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```

```r
median_thresh_maxent <- if(!is.null(crop_maxent)) as.numeric(terra::global(crop_maxent, fun=
thresh_maxent_p88    <- if(!is.null(crop_maxent)) as.numeric(terra::global(crop_maxent, fun=
thresh_maxent_p95    <- if(!is.null(crop_maxent)) as.numeric(terra::global(crop_maxent, fun=

# Calculate P88 and P95 Thresholds
thresh_2025_p88 <- as.numeric(terra::global(crop_2025, fun=function(x) quantile(x, probs=0.88
thresh_2025_p95 <- as.numeric(terra::global(crop_2025, fun=function(x) quantile(x, probs=0.95
```

```r
# Fixed mapping cutoffs
p_min_2025 <- 0.48
p_max_2025 <- 0.55

cat("Median Threshold (2025):", round(median_thresh_2025, 3), "\n")
```

```
Median Threshold (2025): 0.483
```

```r
cat("Median Threshold (Agnostic):", round(median_thresh_agnostic, 3), "\n")
```

```
Median Threshold (Agnostic): 0.562
```

```r
cat("Median Threshold (MaxEnt):", round(median_thresh_maxent, 3), "\n")
```

```
Median Threshold (MaxEnt): 0.06
```

```r
cat("88th Percentile (MaxEnt):", round(thresh_maxent_p88, 3), "\n")
```

```
88th Percentile (MaxEnt): 0.522
```

```r
cat("95th Percentile (MaxEnt):", round(thresh_maxent_p95, 3), "\n")
```

```
95th Percentile (MaxEnt): 0.677
```

```r
cat("88th Percentile (2025):", round(thresh_2025_p88, 3), "\n")
```

```
88th Percentile (2025): 0.506
```

```r
cat("95th Percentile (2025):", round(thresh_2025_p95, 3), "\n")
```

```
95th Percentile (2025): 0.515
```

```r
cat("Mapping Range (2025):", p_min_2025, "to", p_max_2025, "\n")
```

```
Mapping Range (2025): 0.48 to 0.55
```

1. Model Fitting Performance

This section presents the internal performance metrics generated during the XGBoost model training process (`fit_predict_cots_sdm.R`). The model was trained using 5-fold Spatial Block Cross-Validation (50km blocks) on historical cull dive data.

- **Training (In-Fold)** represents the model's performance when tested on the data it was trained on.
- **Validation (Out-of-Fold)** represents the model's performance on the hold-out spatial blocks, providing a robust estimate of how the model generalizes to new reefs.

```
metrics_clean_file <- "outputs/model_performance_metrics_clean.rds"
metrics_rg_file    <- "outputs/model_performance_metrics_withRG.rds"

if (file.exists(metrics_clean_file)) {
  m_clean <- readRDS(metrics_clean_file)

  # Overall Metrics Comparison if RG exists
  if (file.exists(metrics_rg_file)) {
    m_rg <- readRDS(metrics_rg_file)

    cat("### 1.1 Model Comparison: Clean vs. ReefGuide Layers\n\n")
    comp_df <- bind_rows(
      m_clean$Overall %>% mutate(Model = "Clean (12 Layers)"),
      m_rg$Overall %>% mutate(Model = "ReefGuide (15 Layers)")
    ) %>%
      select(Model, Split, roc_auc, accuracy, sens, spec)

    print(kable(comp_df, digits = 3, caption = "Comparison of Internal Cross-Validation Metri
    cat("\n\n")
  }

  # Overall Metrics Table (Detailed)
  cat("### 1.2 Overall Performance: Training vs Validation (Clean Model)\n\n")
  print(kable(m_clean$Overall, digits = 3, caption = "Detailed Metrics for Clean Model"))
  cat("\n\n")

  # Regional Metrics Table
  cat("### 1.3 Regional Validation Performance (Out-of-Fold)\n\n")
  print(kable(m_clean$By_Zone %>% arrange(ManagementZone), digits = 3, caption = "Clean Model
  cat("\n\n")
} else {
```

```
  cat("*Model performance metrics file not found. Ensure `fit_predict_cots_sdm.R` has been ru
}
```

1.1 Model Comparison: Clean vs. ReefGuide Layers

Table 1: Comparison of Internal Cross-Validation Metrics

Model	Split	roc_auc	accuracy	sens	spec
Clean (12 Layers)	Training (In-Fold)	0.740	0.678	0.463	0.857
Clean (12 Layers)	Validation (Out-of-Fold)	0.613	0.583	0.343	0.784
ReefGuide (15 Layers)	Training (In-Fold)	0.967	0.902	0.878	0.924
ReefGuide (15 Layers)	Validation (Out-of-Fold)	0.698	0.651	0.528	0.759

1.2 Overall Performance: Training vs Validation (Clean Model)

Table 2: Detailed Metrics for Clean Model

Split	accuracy	sens	spec	roc_auc
Training (In-Fold)	0.678	0.463	0.857	0.740
Validation (Out-of-Fold)	0.583	0.343	0.784	0.613

1.3 Regional Validation Performance (Out-of-Fold)

Table 3: Clean Model Performance by Management Zone

ManagementZone	accuracy	sens	spec	roc_auc
Central	0.535	0.193	0.844	0.593
Far Northern	0.539	0.108	0.692	0.291
Northern	0.629	0.552	0.694	0.680
Southern	0.571	0.200	0.861	0.576

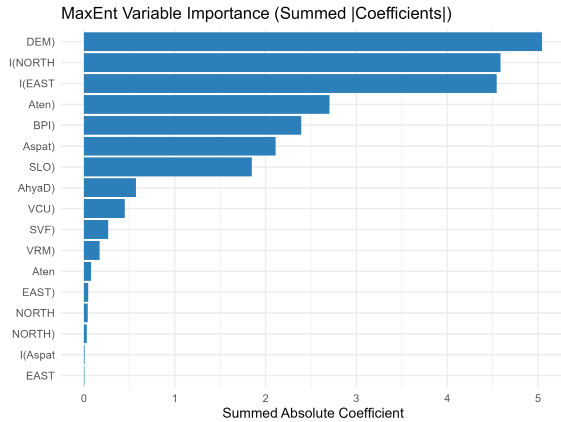
1.4 MaxEnt Presence-Only Model (ENMeval Tuning & Feature Importance)

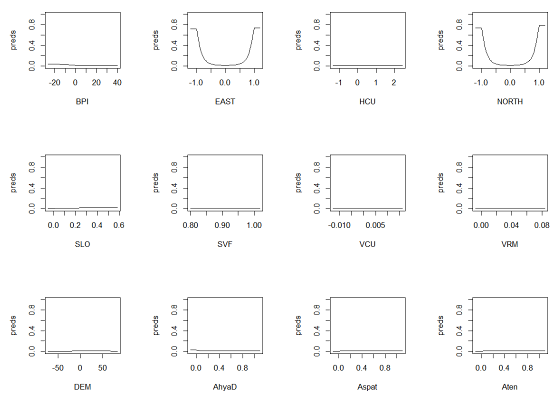
```r
maxent_rds <- "outputs/maxent_enmeval_results.rds"
if (file.exists(maxent_rds)) {
  me_res <- readRDS(maxent_rds)
  res_df <- me_res@results %>%
    select(fc, rm, auc.train, auc.val.avg, auc.val.sd, AICc) %>%
    arrange(AICc)

  best_row <- res_df[1, ]
  cat(paste0("**Optimal MaxEnt Configuration selected by AICc**:\n",
             "- **Feature Classes (fc)**: ", best_row$fc, "\n",
             "- **Regularization Multiplier (rm)**: ", best_row$rm, "\n",
             "- **AICc**: ", round(best_row$AICc, 2), "\n",
             "- **Validation AUC (4-fold Spatial Block CV)**: ", round(best_row$auc.val.avg,

  print(kable(head(res_df, 10), digits = 3, caption = "Top 10 MaxEnt Candidate Model Configu
  cat("\n\n")
} else {
  cat("*MaxEnt ENMeval tuning results not found.*")
}
```

Optimal MaxEnt Configuration selected by AICc: - **Feature Classes (fc)**: LQH - **Regularization Multiplier (rm)**: 0.5 - **AICc**: 139978.57 - **Validation AUC (4-fold Spatial Block CV)**: 0.945 ± 0.023

Table 4: Top 10 MaxEnt Candidate Model Configurations (ENMeval)

fc	rm	auc.train	auc.val.avg	auc.val.sd	AICc
LQH	0.5	0.968	0.945	0.023	139978.6
LQH	1.0	0.968	0.945	0.024	140001.2
LQH	2.0	0.968	0.944	0.025	140065.2
H	0.5	0.968	0.945	0.024	140140.1
LQH	3.0	0.967	0.944	0.025	140143.2
H	1.0	0.967	0.944	0.025	140190.9
LQH	5.0	0.966	0.942	0.026	140355.4
H	2.0	0.966	0.942	0.026	140396.6
H	3.0	0.964	0.940	0.026	140605.5
H	5.0	0.959	0.936	0.028	141078.0





2. Step A: Grid-Based Historical Validation (100m)

We rasterize historical manta tow observations onto a 100m grid and compare them to the SDM probability values (using both 2025 and Agnostic models). We calculate ROC curves, AUC, and evaluate thresholds using confusion matrices.

```
if(nrow(manta_selected) > 0) {
  # crop_2025 is already 60m. Resample crop_agnostic to match it exactly.
  pred_60m_2025 <- crop_2025
  pred_60m_agn <- terra::resample(crop_agnostic, crop_2025, method = "bilinear")
  pred_60m_max <- if(!is.null(crop_maxent)) terra::resample(crop_maxent, crop_2025, method =

  # Rasterize Manta Tow presence (1 if ever present, 0 if absent)
  manta_presence <- terra::rasterize(
    terra::vect(manta_selected),
    pred_60m_2025,
```

```
  field = "presence_any",
  fun = "max",
  background = NA
)

# Build evaluation dataframe
val_df <- data.frame(
  observed_presence = terra::values(manta_presence)[,1],
  prob_2025 = terra::values(pred_60m_2025)[,1],
  prob_agnostic = terra::values(pred_60m_agn)[,1],
  prob_maxent   = if(!is.null(pred_60m_max)) terra::values(pred_60m_max)[,1] else NA
) %>% drop_na()

# --- ROC and AUC ---
roc_2025 <- roc(val_df$observed_presence, val_df$prob_2025, quiet=TRUE)
roc_agn <- roc(val_df$observed_presence, val_df$prob_agnostic, quiet=TRUE)
roc_maxent <- if("prob_maxent" %in% names(val_df)) roc(val_df$observed_presence, val_df$pro

cat("AUC (2025 Model):", round(auc(roc_2025), 3), "\n")
cat("AUC (Agnostic Model):", round(auc(roc_agn), 3), "\n\n")
if(!is.null(roc_maxent)) cat("AUC (MaxEnt Model):", round(auc(roc_maxent), 3), "\n")

# Plot ROC
par(mfrow=c(1,2))
plot(roc_2025, main="ROC Curve (2025)", col="blue", lwd=2)
plot(roc_agn, main="ROC Curve (Agnostic)", col="red", lwd=2)

# --- Confusion Matrix at Median Threshold ---
cat("Confusion Matrix for 2025 Model (Threshold = Median:", round(median_thresh_2025,3), ")
cm_2025 <- table(
  Predicted = factor(val_df$prob_2025 >= median_thresh_2025, levels=c(FALSE, TRUE)),
  Observed = factor(val_df$observed_presence == 1, levels=c(FALSE, TRUE))
)
print(cm_2025)
cat("Accuracy:", sum(diag(cm_2025))/sum(cm_2025), "\n\n")

cat("Confusion Matrix for Agnostic Model (Threshold = Median:", round(median_thresh_agnosti
cm_agn <- table(
  Predicted = factor(val_df$prob_agnostic >= median_thresh_agnostic, levels=c(FALSE, TRUE))
  Observed = factor(val_df$observed_presence == 1, levels=c(FALSE, TRUE))
)
print(cm_agn)
```

```
  cat("Accuracy:", sum(diag(cm_agn))/sum(cm_agn), "\n")
}
```

```
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AUC (2025 Model): 0.506
AUC (Agnostic Model): 0.602

AUC (MaxEnt Model): 0.642
```
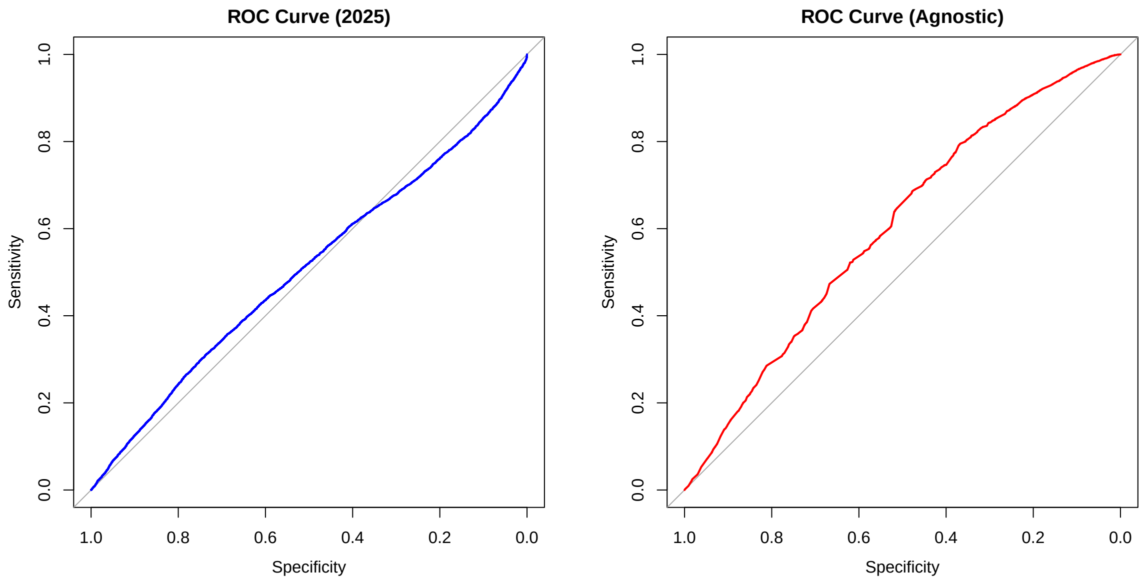




```
Confusion Matrix for 2025 Model (Threshold = Median: 0.483 )
         Observed
Predicted FALSE  TRUE
    FALSE 60694  5241
    TRUE  27787  2588
Accuracy: 0.6570657

Confusion Matrix for Agnostic Model (Threshold = Median: 0.562 )
         Observed
Predicted FALSE  TRUE
    FALSE 60085  6499
    TRUE  28396  1330
Accuracy: 0.6376804
```

2b. Cull Dive Validation (ROC & AUC)

Cull dive data provides a more reliable validation target than manta tow data because divers directly count COTS at each site. We extract the SDM probability at each cull dive location and evaluate model discrimination using ROC curves. Presence is defined as `Total > 0` (at least one COTS found). We use all cull dive data from Jan 2025 onwards for the top 30 reefs.

```
# Extract SDM probabilities at cull dive point locations
cull_pts_2025 <- st_as_sf(
  cull_raw %>% filter(ReefName %in% reef_select_eval, as.Date(SurveyDate) >= as.Date("2025-0
```

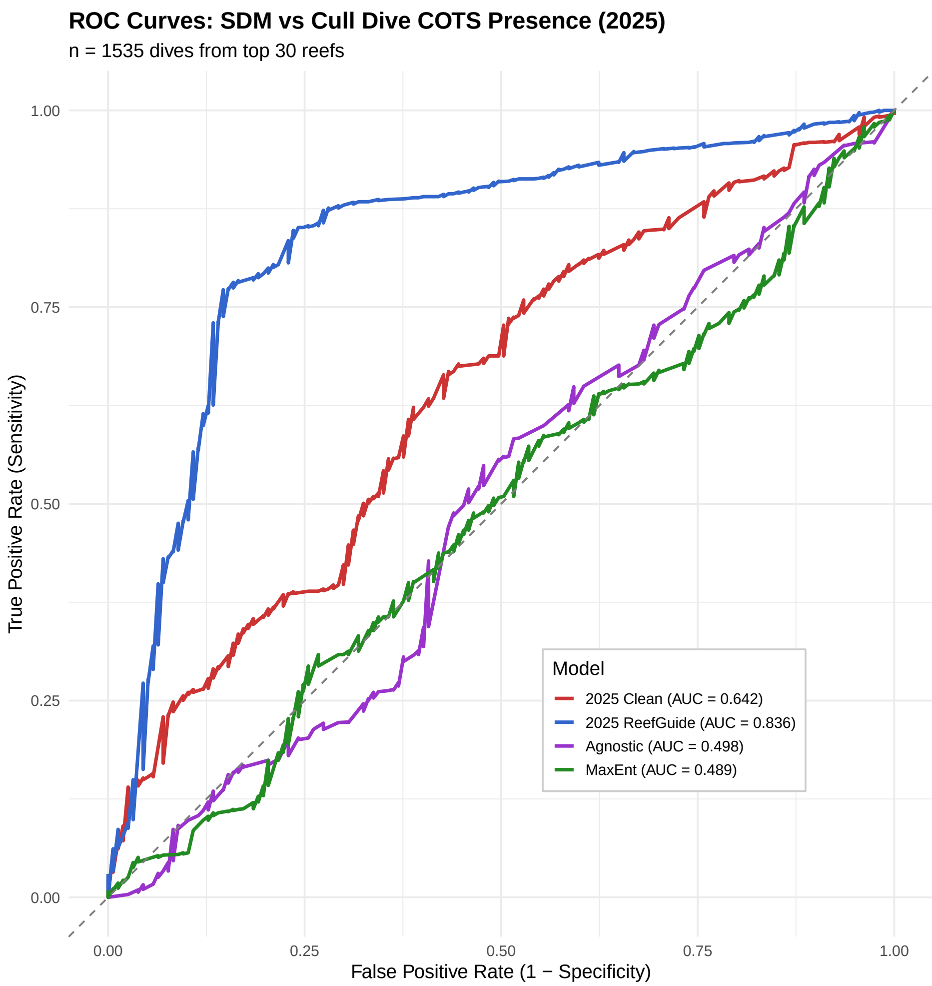
```
  coords = c("Longitude", "Latitude"), crs = 4326
) %>% st_transform(crs = crs(prob_year2025))

# Extract 2025 model probability
cull_extract_2025 <- terra::extract(prob_year2025, terra::vect(cull_pts_2025))
val_col_2025 <- setdiff(names(cull_extract_2025), "ID")[1]

# Extract agnostic model probability
cull_extract_agn <- terra::extract(prob_agnostic, terra::vect(cull_pts_2025))
# Extract MaxEnt model probability
cull_extract_max <- if(!is.null(prob_maxent)) terra::extract(prob_maxent, terra::vect(cull_pt
val_col_max <- if(!is.null(cull_extract_max)) setdiff(names(cull_extract_max), "ID")[1] else
val_col_agn <- setdiff(names(cull_extract_agn), "ID")[1]

# Extract RG model probability (if available)
cull_extract_rg <- if(!is.null(prob_2025_rg)) terra::extract(prob_2025_rg, terra::vect(cull_p
val_col_rg <- if(!is.null(cull_extract_rg)) setdiff(names(cull_extract_rg), "ID")[1] else NUL

# Build validation dataframe
cull_val_df <- data.frame(
  observed_presence = as.integer(cull_pts_2025$Total > 0),
  prob_2025 = cull_extract_2025[[val_col_2025]],
  prob_agnostic = cull_extract_agn[[val_col_agn]],
  prob_maxent   = if(!is.null(cull_extract_max)) cull_extract_max[[val_col_max]] else NA
)

if (!is.null(cull_extract_rg)) {
  cull_val_df$prob_rg <- cull_extract_rg[[val_col_rg]]
}

cull_val_df <- cull_val_df %>%
  bind_cols(cull_pts_2025 %>% st_drop_geometry() %>% select(Total, CPUE)) %>%
  drop_na()

cat("Cull dive validation dataset:", nrow(cull_val_df), "dives\n")
```

```
Cull dive validation dataset: 1535 dives
```

```
# ... (Counts output) ...

# --- ROC Curves ---
```

```r
roc_cull_2025 <- roc(cull_val_df$observed_presence, cull_val_df$prob_2025, quiet = TRUE)
roc_cull_agn  <- roc(cull_val_df$observed_presence, cull_val_df$prob_agnostic, quiet = TRUE)
roc_cull_max  <- if("prob_maxent" %in% names(cull_val_df)) roc(cull_val_df$observed_presence
roc_cull_rg   <- if("prob_rg" %in% names(cull_val_df)) roc(cull_val_df$observed_presence, cul

# Build ROC dataframes for ggplot
roc_df_2025 <- data.frame(
  FPR = 1 - roc_cull_2025$specificities,
  TPR = roc_cull_2025$sensitivities,
  Model = paste0("2025 Clean (AUC = ", round(auc(roc_cull_2025), 3), ")")
)
roc_df_agn <- data.frame(
  FPR = 1 - roc_cull_agn$specificities,
  TPR = roc_cull_agn$sensitivities,
  Model = paste0("Agnostic (AUC = ", round(auc(roc_cull_agn), 3), ")")
)
roc_df_max <- if(!is.null(roc_cull_max)) data.frame(
  FPR = 1 - roc_cull_max$specificities,
  TPR = roc_cull_max$sensitivities,
  Model = paste0("MaxEnt (AUC = ", round(auc(roc_cull_max), 3), ")")
) else NULL
roc_df <- bind_rows(roc_df_2025, roc_df_agn, roc_df_max)

if (!is.null(roc_cull_rg)) {
  roc_df_rg <- data.frame(
    FPR = 1 - roc_cull_rg$specificities,
    TPR = roc_cull_rg$sensitivities,
    Model = paste0("2025 ReefGuide (AUC = ", round(auc(roc_cull_rg), 3), ")")
  )
  roc_df <- bind_rows(roc_df, roc_df_rg)
}

# ggplot ROC curves
p_roc <- ggplot(roc_df, aes(x = FPR, y = TPR, color = Model)) +
  geom_line(linewidth = 1.2) +
  geom_abline(linetype = "dashed", color = "grey50") +
  scale_color_manual(values = c("#cc3333", "#3366cc", "#9933cc", "#228b22", "#ff7f00")) +
  theme_minimal(base_size = 14) +
  labs(
    title = "ROC Curves: SDM vs Cull Dive COTS Presence (2025)",
    subtitle = paste0("n = ", nrow(cull_val_df), " dives from top 30 reefs"),
    x = "False Positive Rate (1 - Specificity)",
```

```
    y = "True Positive Rate (Sensitivity)",
    color = "Model"
  ) +
  theme(
    plot.title = element_text(face = "bold"),
    legend.position = c(0.7, 0.25),
    legend.background = element_rect(fill = "white", color = "grey80")
  ) +
  coord_equal()

print(p_roc)
```



```r
# --- Optimal thresholds (Youden's J) ---
opt_2025 <- coords(roc_cull_2025, "best", best.method = "youden", ret = c("threshold", "sensi
opt_agn  <- coords(roc_cull_agn,  "best", best.method = "youden", ret = c("threshold", "sensi

cat("Optimal Threshold (Youden's J) for 2025 Model:", round(opt_2025$threshold, 4),
    "  Sensitivity:", round(opt_2025$sensitivity, 3),
    "  Specificity:", round(opt_2025$specificity, 3), "\n")
```

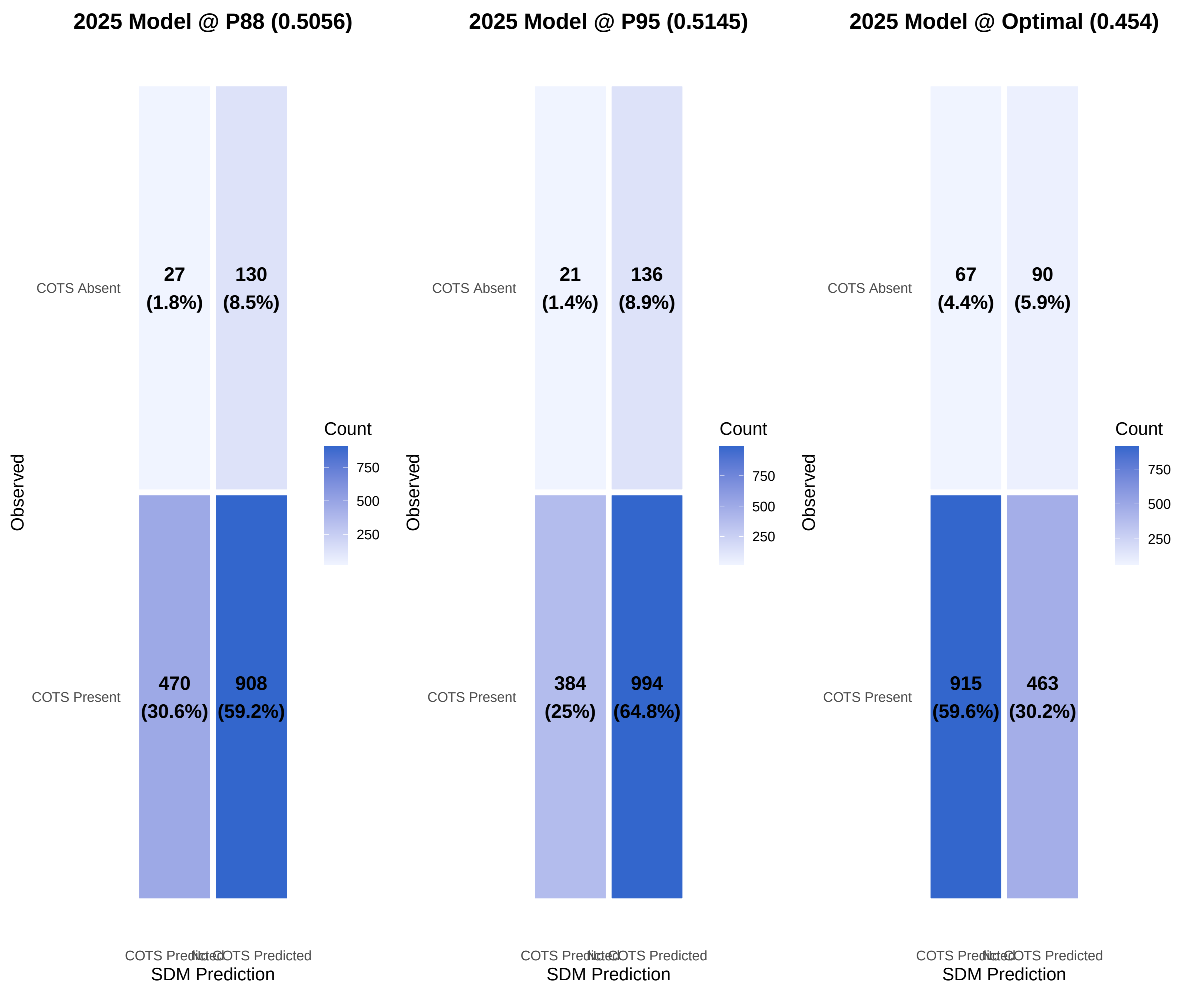
```
Optimal Threshold (Youden's J) for 2025 Model: 0.454   Sensitivity: 0.664   Specificity: 0.5
```

```r
cat("Optimal Threshold (Youden's J) for Agnostic Model:", round(opt_agn$threshold, 4),
    "  Sensitivity:", round(opt_agn$sensitivity, 3),
    "  Specificity:", round(opt_agn$specificity, 3), "\n\n")
```

```
Optimal Threshold (Youden's J) for Agnostic Model: 0.5227   Sensitivity: 0.549   Specificity
```

```r
# --- Confusion Matrices at P88 and P95 thresholds ---
plot_cull_cm <- function(prob_vec, obs_vec, threshold, title) {
  cm_df <- data.frame(
    Predicted = factor(ifelse(prob_vec >= threshold, "COTS Predicted", "No COTS Predicted"),
                       levels = c("COTS Predicted", "No COTS Predicted")),
    Observed  = factor(ifelse(obs_vec, "COTS Present", "COTS Absent"),
                       levels = c("COTS Present", "COTS Absent"))
  )
  cm_counts <- cm_df %>%
    count(Predicted, Observed) %>%
    group_by() %>%
    mutate(pct = n / sum(n) * 100)

  ggplot(cm_counts, aes(x = Predicted, y = Observed, fill = n)) +
    geom_tile(color = "white", linewidth = 2) +
    geom_text(aes(label = paste0(n, "\n(", round(pct, 1), "%)")), size = 5, fontface = "bold"
    scale_fill_gradient(low = "#f0f4ff", high = "#3366cc") +
    theme_minimal(base_size = 13) +
    labs(title = title, x = "SDM Prediction", y = "Observed", fill = "Count") +
    theme(plot.title = element_text(face = "bold", hjust = 0.5), panel.grid = element_blank()
}

p_cm1 <- plot_cull_cm(cull_val_df$prob_2025, cull_val_df$observed_presence == 1,
                      thresh_2025_p88, paste0("2025 Model @ P88 (", round(thresh_2025_p88, 4
```

```
p_cm2 <- plot_cull_cm(cull_val_df$prob_2025, cull_val_df$observed_presence == 1,
                       thresh_2025_p95, paste0("2025 Model @ P95 (", round(thresh_2025_p95, 4
p_cm3 <- plot_cull_cm(cull_val_df$prob_2025, cull_val_df$observed_presence == 1,
                       opt_2025$threshold, paste0("2025 Model @ Optimal (", round(opt_2025$tl

gridExtra::grid.arrange(p_cm1, p_cm2, p_cm3, ncol = 3)
```



3. Step B: Tow-Path Simulation and Evaluation

Simulating whether a specific tow path would have been suggested by the 2025 SDM model. We evaluate hit-rates strictly against tows conducted from Jan 1 2025 onwards, however, we use all historic tow tracks to delineate the operational boundaries.

```
# 2. Define Regions (100m Buffered Tows & 200m Grids)
tow_regions_list <- list()
grid_regions_list <- list()

for (r_name in unique(manta_eval_all$ReefName)) {
  m_sub <- manta_eval_all %>% filter(ReefName == r_name)
  if (nrow(m_sub) == 0) next

  bb <- st_bbox(m_sub)
  gr_500 <- st_make_grid(bb, cellsize = 500, square = TRUE) %>% st_as_sf()

  # Tow Regions (100m buffer, unioned, then intersected with 500m grid)
  tr <- st_buffer(m_sub, dist = 100) %>%
    st_union() %>%
    st_intersection(gr_500) %>%
    st_cast("POLYGON") %>%
    st_as_sf() %>%
    mutate(ReefName = r_name, RegionType = "Tow_100m_Seg500m", RegionID = paste0(r_name, "_T
    rename(geometry = x)
  tow_regions_list[[r_name]] <- tr

  # Grid Regions (200m x 200m)
  gr <- st_make_grid(bb, cellsize = 200, square = TRUE) %>%
    st_as_sf() %>%
    mutate(ReefName = r_name, RegionType = "Grid_200m", RegionID = paste0(r_name, "_G_", row
    rename(geometry = x)

  # Keep only grid cells that intersect with manta tows to avoid evaluating empty ocean
  gr <- gr[lengths(st_intersects(gr, m_sub)) > 0, ]

  grid_regions_list[[r_name]] <- gr
}

tow_regions <- bind_rows(tow_regions_list)
grid_regions <- bind_rows(grid_regions_list)
all_regions <- bind_rows(tow_regions, grid_regions)

# 3. Extract SDM Stats for Regions
extract_poly_stats <- function(r, polys) {
  v <- terra::vect(polys)
  ex <- terra::extract(r, v)
  val_col <- setdiff(names(ex), "ID")[1]
```

```
  ex %>%
    rename(poly_id = ID, value = !!val_col) %>%
    group_by(poly_id) %>%
    summarise(
      reg_p88    = as.numeric(quantile(value, probs = 0.88, na.rm = TRUE, names = FALSE)),
      reg_p95    = as.numeric(quantile(value, probs = 0.95, na.rm = TRUE, names = FALSE)),
      .groups    = "drop"
    )
}

all_regions$poly_id <- seq_len(nrow(all_regions))
region_stats <- extract_poly_stats(prob_year2025, all_regions)

all_regions <- all_regions %>% left_join(region_stats, by = "poly_id")
tow_regions <- all_regions %>% filter(RegionType == "Tow_100m_Seg500m")
grid_regions <- all_regions %>% filter(RegionType == "Grid_200m")

# 4. Map Tows to Regions and Apply Thresholds
# using st_join(largest = TRUE) ensures a tow is assigned to only one region
m_tow <- st_join(manta_eval_all, tow_regions %>% select(RegionID_Tow = RegionID, reg_p88_Tow
m_grid <- st_join(manta_eval_all, grid_regions %>% select(RegionID_Grid = RegionID, reg_p88_

manta_eval <- manta_eval_all %>%
  mutate(
    reg_p88_Tow = m_tow$reg_p88_Tow,
    reg_p95_Tow = m_tow$reg_p95_Tow,
    reg_p88_Grid = m_grid$reg_p88_Grid,
    reg_p95_Grid = m_grid$reg_p95_Grid
  ) %>%
  mutate(
    sug_Tow_P88 = !is.na(reg_p88_Tow) & reg_p88_Tow >= thresh_2025_p88,
    sug_Tow_P95 = !is.na(reg_p95_Tow) & reg_p95_Tow >= thresh_2025_p95,
    sug_Grid_P88 = !is.na(reg_p88_Grid) & reg_p88_Grid >= thresh_2025_p88,
    sug_Grid_P95 = !is.na(reg_p95_Grid) & reg_p95_Grid >= thresh_2025_p95
  )

# Function to compute hit rates by reef for a specific scenario
calc_hit_rates_reef <- function(df, scenario_col, scenario_name) {
  overall <- df %>%
    summarise(
      ReefName = "Overall",
      Scenario = scenario_name,
```

```
      Total_Tows = n(),
      Tows_Suggested = sum(!!sym(scenario_col), na.rm=TRUE),
      Hit_Rate = sum(!!sym(scenario_col) & presence_any, na.rm=TRUE) / Tows_Suggested,
      Tows_Avoided = sum(! (!!sym(scenario_col)), na.rm=TRUE),
      Miss_Rate = sum(! (!!sym(scenario_col)) & presence_any, na.rm=TRUE) / Tows_Avoided
    )
  by_reef <- df %>%
    group_by(ReefName) %>%
    summarise(
      Scenario = scenario_name,
      Total_Tows = n(),
      Tows_Suggested = sum(!!sym(scenario_col), na.rm=TRUE),
      Hit_Rate = sum(!!sym(scenario_col) & presence_any, na.rm=TRUE) / Tows_Suggested,
      Tows_Avoided = sum(! (!!sym(scenario_col)), na.rm=TRUE),
      Miss_Rate = sum(! (!!sym(scenario_col)) & presence_any, na.rm=TRUE) / Tows_Avoided
    )
  bind_rows(by_reef, overall)
}

manta_eval_2025 <- manta_eval %>% filter(as.Date(SurveyTime) >= as.Date("2025-01-01"))

hit_table <- bind_rows(
  calc_hit_rates_reef(manta_eval_2025 %>% filter(ReefName %in% reef_select), "sug_Tow_P88",
  calc_hit_rates_reef(manta_eval_2025 %>% filter(ReefName %in% reef_select), "sug_Tow_P95",
) %>% arrange(ReefName, desc(Scenario))

kable(hit_table, digits = 3, caption = "Manta Tow Simulation: Operational Zonal Validation (
```

Table 5: Manta Tow Simulation: Operational Zonal Validation (Selected 10 Reefs)

ReefName	Scenario	Total_Tows	Tows_Suggested	Hit_Rate	Tows_Avoided	Miss_Rate	geometry
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P95	637	332	0.099	305	0.128	MULTILINESTRING ((573577.7 ...
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P88	637	185	0.076	452	0.128	MULTILINESTRING ((573577.7 ...
Batt Reef (16-029)	Seg Tow 100m + P95	382	344	0.308	38	0.026	MULTILINESTRING ((377707.6 ...

ReefName	Scenario	Total_Tows	Tows_Suggested	Hit_Rate	Tows_Avoided	Miss_Rate	geometry
Batt Reef (16-029)	Seg Tow 100m + P88	382	317	0.312	65	0.123	MULTILINESTRING ((377707.6 ...
Davies Reef (18-096)	Seg Tow 100m + P95	177	100	0.200	77	0.260	MULTILINESTRING ((566203.3 ...
Davies Reef (18-096)	Seg Tow 100m + P88	177	60	0.117	117	0.282	MULTILINESTRING ((566203.3 ...
Heron Reef (23-052a)	Seg Tow 100m + P95	217	163	0.147	54	0.093	MULTILINESTRING ((1003623 7...
Heron Reef (23-052a)	Seg Tow 100m + P88	217	154	0.149	63	0.095	MULTILINESTRING ((1003623 7...
Overall	Seg Tow 100m + P95	1413	939	0.195	474	0.137	MULTILINESTRING ((377707.6 ...
Overall	Seg Tow 100m + P88	1413	716	0.200	697	0.151	MULTILINESTRING ((377707.6 ...

4. Step C: Cull Dive Financial Savings

Evaluating cull dives strictly from Jan 1 2025 onwards using the 2025 P88 and P95 thresholds on Segmented Tow Regions. We quantify avoided diving hours for dives that resulted in 0.00 CPUE (Ghost Culls) and <0.02 CPUE (Non-Problematic), and calculate financial savings ($950/hr).

```
# --- Step 1: Build cull geometry using polygons where available, points otherwise ---
# Join cull data to polygon geometries by CullSiteName
cull_eval_raw <- cull_raw %>% filter(ReefName %in% reef_select_eval)
cull_eval_raw$row_id <- seq_len(nrow(cull_eval_raw))

# Match cull site names to polygon geometries
poly_lookup <- cull_site_polys %>% select(CullSiteName)

matched_idx <- cull_eval_raw$CullSiteName %in% poly_lookup$CullSiteName
cat("Cull site polygon match rate:", round(mean(matched_idx) * 100, 1), "%\n")
```

```
Cull site polygon match rate: 100 %
```

```r
cat("  Matched:", sum(matched_idx), "  Unmatched:", sum(!matched_idx), "\n")
```

```
  Matched: 9426   Unmatched: 0
```

```r
# For matched: use polygon + 50m buffer
if (sum(matched_idx) > 0) {
  # Look up the polygon for each matched cull record
  match_positions <- match(cull_eval_raw$CullSiteName[matched_idx], poly_lookup$CullSiteName)
  matched_geoms <- st_buffer(st_geometry(poly_lookup)[match_positions], dist = 50)
  cull_matched_sf2 <- st_sf(cull_eval_raw[matched_idx, ], geometry = matched_geoms)
  cull_matched_sf2$match_type <- "Polygon"
} else {
  cull_matched_sf2 <- NULL
}

# For unmatched: use point + 100m buffer
if (sum(!matched_idx) > 0) {
  cull_unmatched_pts <- st_as_sf(cull_eval_raw[!matched_idx, ],
    coords = c("Longitude", "Latitude"), crs = 4326) %>%
    st_transform(crs = crs(prob_year2025))
  cull_unmatched_buf <- st_buffer(cull_unmatched_pts, dist = 100)
  cull_unmatched_buf$match_type <- "Point (100m buffer)"
} else {
  cull_unmatched_buf <- NULL
}

# Combine into single sf
cull_sf_combined <- bind_rows(cull_matched_sf2, cull_unmatched_buf) %>%
  arrange(row_id)

cat("Combined cull geometries:", nrow(cull_sf_combined), "rows\n")
```

```
Combined cull geometries: 9426 rows
```

```r
# --- Step 2: Spatial join to tow/grid regions ---
c_tow  <- st_join(cull_sf_combined, tow_regions  %>% select(RegionID_Tow  = RegionID, reg_p8
c_grid <- st_join(cull_sf_combined, grid_regions %>% select(RegionID_Grid = RegionID, reg_p8

# --- Step 3: Sector join (using centroids for sector assignment) ---
```

```r
sectors_sf <- st_read("data/SectorShapefile/ltms.shp", quiet = TRUE) %>%
  st_transform(crs = crs(prob_year2025))
cull_centroids <- st_centroid(cull_sf_combined)
c_sector <- st_join(cull_centroids, sectors_sf %>% select(SECTOR_NAM), join = st_nearest_fea

# --- Step 4: For unmatched tow-region NAs, extract MEAN raster value across the cull polygo
na_tow_idx <- is.na(c_tow$reg_p88_Tow)
cat("Tow region NA count:", sum(na_tow_idx), "of", nrow(c_tow), "\n")
```

```
Tow region NA count: 12 of 9426
```

```r
# Extract mean raster value for each cull site polygon/buffer that didn't match a tow region
if (sum(na_tow_idx) > 0) {
  na_polys <- cull_sf_combined[na_tow_idx, ]
  na_extract <- terra::extract(prob_year2025, terra::vect(na_polys), fun = "mean", na.rm = T
  val_col <- setdiff(names(na_extract), "ID")[1]
  inferred_vals <- na_extract[[val_col]]
} else {
  inferred_vals <- numeric(0)
}

# --- Step 5: Build cull_eval with combined logic ---
cull_eval <- cull_sf_combined %>%
  st_drop_geometry() %>%
  mutate(
    reg_p88_Tow  = c_tow$reg_p88_Tow,
    reg_p95_Tow  = c_tow$reg_p95_Tow,
    reg_p88_Grid = c_grid$reg_p88_Grid,
    reg_p95_Grid = c_grid$reg_p95_Grid,
    SECTOR_NAM   = c_sector$SECTOR_NAM
  )

# For NAs: fill in with inferred raster mean value
cull_eval$inferred_mean <- NA_real_
cull_eval$inferred_mean[na_tow_idx] <- inferred_vals

cull_eval <- cull_eval %>%
  mutate(
    # Region-matched classification
    avoid_Tow_P88 = case_when(
      !is.na(reg_p88_Tow) ~ reg_p88_Tow < thresh_2025_p88,
```

```
      TRUE                  ~ inferred_mean < thresh_2025_p88
    ),
    avoid_Tow_P95 = case_when(
      !is.na(reg_p95_Tow) ~ reg_p95_Tow < thresh_2025_p95,
      TRUE                  ~ inferred_mean < thresh_2025_p95
    ),
    avoid_Grid_P88 = case_when(
      !is.na(reg_p88_Grid) ~ reg_p88_Grid < thresh_2025_p88,
      TRUE                   ~ inferred_mean < thresh_2025_p88
    ),
    avoid_Grid_P95 = case_when(
      !is.na(reg_p95_Grid) ~ reg_p95_Grid < thresh_2025_p95,
      TRUE                   ~ inferred_mean < thresh_2025_p95
    ),
    # Transparency flag
    classification_type = case_when(
      !is.na(reg_p88_Tow) ~ "Region Matched",
      !is.na(inferred_mean) ~ "Inferred",
      TRUE ~ "Unknown"
    )
  )

cat("\nClassification type breakdown:\n")
```

```
Classification type breakdown:
```

```
print(table(cull_eval$classification_type))
```

```
      Inferred Region Matched 
            12           9414 
```

```
diver_hour_cost <- 950

calc_savings_reef <- function(df, avoid_col, scenario_name) {
  overall <- df %>%
    summarise(
      ReefName = "Overall",
      Scenario = scenario_name,
```

```
      Ghost_Avoided = sum(is_ghost & !!sym(avoid_col)),
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Avoided = sum(is_non_problematic & !!sym(avoid_col)),
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_h
      Total_Savings = Ghost_Savings + NonProb_Savings,
      Prob_Missed = sum(!is_non_problematic & !!sym(avoid_col)),
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
      Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60
    ) %>%
    mutate(
      CPUE_Missed = ifelse(Hours_Missed > 0, COTS_Missed / (Hours_Missed * 60), 0)
    )
  by_reef <- df %>%
    group_by(ReefName) %>%
    summarise(
      Scenario = scenario_name,
      Ghost_Avoided = sum(is_ghost & !!sym(avoid_col)),
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Avoided = sum(is_non_problematic & !!sym(avoid_col)),
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_h
      Total_Savings = Ghost_Savings + NonProb_Savings,
      Prob_Missed = sum(!is_non_problematic & !!sym(avoid_col)),
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
      Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60
    ) %>%
    mutate(
      CPUE_Missed = ifelse(Hours_Missed > 0, COTS_Missed / (Hours_Missed * 60), 0)
    )
  bind_rows(by_reef, overall)
}

cull_eval_2025 <- cull_eval %>%
  filter(as.Date(SurveyDate) >= as.Date("2025-01-01")) %>%
  mutate(
    SECTOR_NAM = as.character(SECTOR_NAM),
    ManagementZone = case_when(
      SECTOR_NAM %in% c("Cape Grenville", "Princess Charlotte Bay") ~ "Far Northern",
      SECTOR_NAM %in% c("Cooktown / Lizard Island", "Cairns", "Innisfail") ~ "Northern",
      SECTOR_NAM %in% c("Townsville", "Cape Upstart", "Whitsunday") ~ "Central",
      SECTOR_NAM %in% c("Pompey", "Swain", "Capricorn Bunker") ~ "Southern",
```

```r
      TRUE ~ "Unknown"
    )
  )

# Only show reef-level summary for the 10 selected reefs to avoid massive tables
savings_table <- bind_rows(
  calc_savings_reef(cull_eval_2025 %>% filter(ReefName %in% reef_select), "avoid_Tow_P88", "S
  calc_savings_reef(cull_eval_2025 %>% filter(ReefName %in% reef_select), "avoid_Tow_P95", "S
) %>% arrange(ReefName, desc(Scenario))

kable(savings_table, digits = 2, format.args = list(big.mark = ","), caption = "Financial Sav
```

Table 6: Financial Savings and Risk Summary (Selected 10 Reefs)

ReefName	Scenario	Ghost_Avoided	Ghost_Savings	NonProd_Avoided	NonProd_Savings	Total_Savings	Segs_Missed	COTS_Missed	Pct_COTS_Missed	Hours_Missed	CPUE_Missed
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P95	14	55,654.17	31	139,254.17	194,908.33	14	185	16.59	83.33	0.04
(Big) Broadhurst Reef (No 1) (18-100a)	Seg Tow 100m + P88	26	97,897.50	51	211,960.83	309,858.33	24	430	38.57	133.33	0.05
Batt Reef (16-029)	Seg Tow 100m + P95	0	0.00	0	0.00	0.00	0	0	0.00	0.00	0.00
Batt Reef (16-029)	Seg Tow 100m + P88	1	2,533.33	1	2,533.33	5,066.67	14	277	6.29	77.50	0.06
Davies Reef (18-096)	Seg Tow 100m + P95	1	3,483.33	12	39,963.33	43,446.67	13	94	43.52	49.87	0.03
Davies Reef (18-096)	Seg Tow 100m + P88	1	3,483.33	16	52,946.67	56,430.00	17	113	52.31	62.93	0.03

ReefName	Scenario	Ghost_Avoided	Ghost_Savings	NonProb_Avoided	NonProb_Savings	Total_Savings	Sites_Missed	COTS_Missed	Pct_COTS_Missed	Hours_Missed	CPUE_Missed
Heron Reef (23-052a)	Seg Tow 100m + P95	1	5,510.00	6	29,925.00	35,435.00	0	0	0.00	0.00	0.00
Heron Reef (23-052a)	Seg Tow 100m + P88	1	5,510.00	7	34,295.00	39,805.00	0	0	0.00	0.00	0.00
Overall	Seg Tow 100m + P95	16	64,647.50	49	209,142.50	273,790.00	27	279	4.59	133.20	0.03
Overall	Seg Tow 100m + P88	29	109,424.17	75	301,735.83	411,160.00	55	820	13.50	273.77	0.05

```
# Management Zone & Sector Summary (All 128 Reefs)
calc_savings_zone <- function(df, avoid_col, scenario_name) {
  df %>%
    group_by(ManagementZone, SECTOR_NAM) %>%
    summarise(
      Scenario = scenario_name,
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_h
      Total_Savings = Ghost_Savings + NonProb_Savings,
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0),
      Hours_Missed = sum(Bottomtime[!is_non_problematic & !!sym(avoid_col)]) / 60,
      .groups = "drop"
    )
}

savings_zone_table <- bind_rows(
  calc_savings_zone(cull_eval_2025, "avoid_Tow_P88", "Seg Tow 100m + P88"),
  calc_savings_zone(cull_eval_2025, "avoid_Tow_P95", "Seg Tow 100m + P95")
) %>%
  arrange(factor(ManagementZone, levels = c("Far Northern", "Northern", "Central", "Southern"

kable(savings_zone_table, digits = 2, format.args = list(big.mark = ","), caption = "Financia
```

Table 7: Financial Savings and Risk Summary by Management Zone & Sector (Top 30 Reefs, 2025)

ManagementZone	SECTOR_NAME	Scenario	Ghost_Savings	NonProb_Savings	Total_Savings	COTS_Missed	Pct_COTS_Missed	Hours_Missed
Far Northern	Cape Grenville	Seg Tow 100m + P95	9,468.33	42,085.00	51,553.33	428	48.20	94.42
Far Northern	Cape Grenville	Seg Tow 100m + P88	9,468.33	62,747.50	72,215.83	610	68.69	152.88
Northern	Cairns	Seg Tow 100m + P95	26,916.67	32,458.33	59,375.00	0	0.00	0.00
Northern	Cairns	Seg Tow 100m + P88	29,450.00	70,125.83	99,575.83	588	10.55	184.33
Northern	Cooktown / Lizard Island	Seg Tow 100m + P95	490.83	23,607.50	24,098.33	663	7.13	133.93
Northern	Cooktown / Lizard Island	Seg Tow 100m + P88	490.83	53,849.17	54,340.00	1,191	12.82	266.75
Northern	Innisfail	Seg Tow 100m + P95	0.00	0.00	0.00	0	0.00	0.00
Northern	Innisfail	Seg Tow 100m + P88	6,016.67	37,366.67	43,383.33	25	2.21	13.33
Central	Cape Upstart	Seg Tow 100m + P95	34,643.33	63,903.33	98,546.67	22	13.25	10.17
Central	Cape Upstart	Seg Tow 100m + P88	66,373.33	105,386.67	171,760.00	68	40.96	25.17
Central	Townsville	Seg Tow 100m + P95	65,945.83	353,669.17	419,615.00	647	10.01	290.83
Central	Townsville	Seg Tow 100m + P88	116,881.67	494,601.67	611,483.33	1,230	19.04	419.53

Management_Zone	SECTOR_NAME	Scenario	Ghost_Savings	NonProb_Savings	Total_Savings	COTS_Missed	Pct_COTS_Missed	Hours_Missed
Southern	Capricorn Bunker	Seg Tow 100m + P95	28,785.00	147,535.00	176,320.00	664	14.66	129.88
Southern	Capricorn Bunker	Seg Tow 100m + P88	31,207.50	174,356.67	205,564.17	934	20.62	188.78
Southern	Swain	Seg Tow 100m + P95	46,122.50	107,255.00	153,377.50	487	20.51	94.88
Southern	Swain	Seg Tow 100m + P88	61,180.00	145,967.50	207,147.50	696	29.31	159.20

```
# --- Consolidated Scenario Summary ---
scenario_summary <- function(manta_df, cull_df, tow_col, avoid_col, label) {
  # Manta tow summary
  m <- manta_df %>%
    summarise(
      Total_Tows = n(),
      Tows_Suggested = sum(!!sym(tow_col), na.rm=TRUE),
      Tows_Avoided = Total_Tows - Tows_Suggested,
      Hit_Rate = sum(!!sym(tow_col) & presence_any, na.rm=TRUE) / max(Tows_Suggested, 1),
      Miss_Rate = sum(!(!!sym(tow_col)) & presence_any, na.rm=TRUE) / max(Tows_Avoided, 1)
    )
  # Cull dive summary
  c <- cull_df %>%
    summarise(
      Total_Dives = n(),
      Dives_Avoided = sum(!!sym(avoid_col), na.rm=TRUE),
      Ghost_Savings = sum(Bottomtime[is_ghost & !!sym(avoid_col)]) / 60 * diver_hour_cost,
      NonProb_Savings = sum(Bottomtime[is_non_problematic & !!sym(avoid_col)]) / 60 * diver_
      Total_Savings = Ghost_Savings + NonProb_Savings,
      COTS_Missed = sum(Total[!is_non_problematic & !!sym(avoid_col)]),
      Pct_COTS_Missed = ifelse(sum(Total) > 0, COTS_Missed / sum(Total) * 100, 0)
    )
  tibble(
    Scenario = label,
    Total_Tows = m$Total_Tows, Tows_Suggested = m$Tows_Suggested, Tows_Avoided = m$Tows_Avoi
    Tow_Hit_Rate = m$Hit_Rate, Tow_Miss_Rate = m$Miss_Rate,
    Total_Dives = c$Total_Dives, Dives_Avoided = c$Dives_Avoided,
```

```
    Total_Savings = c$Total_Savings, COTS_Missed = c$COTS_Missed, Pct_COTS_Missed = c$Pct_CO
  )
}

consolidated <- bind_rows(
  scenario_summary(manta_eval_2025, cull_eval_2025, "sug_Tow_P88", "avoid_Tow_P88", "P88 (Ag
  scenario_summary(manta_eval_2025, cull_eval_2025, "sug_Tow_P95", "avoid_Tow_P95", "P95 (Co
)

kable(consolidated, digits = 2, format.args = list(big.mark = ","),
      caption = "Consolidated Scenario Comparison: Manta Tow + Cull Dive (All Top 30 Reefs, 2
```

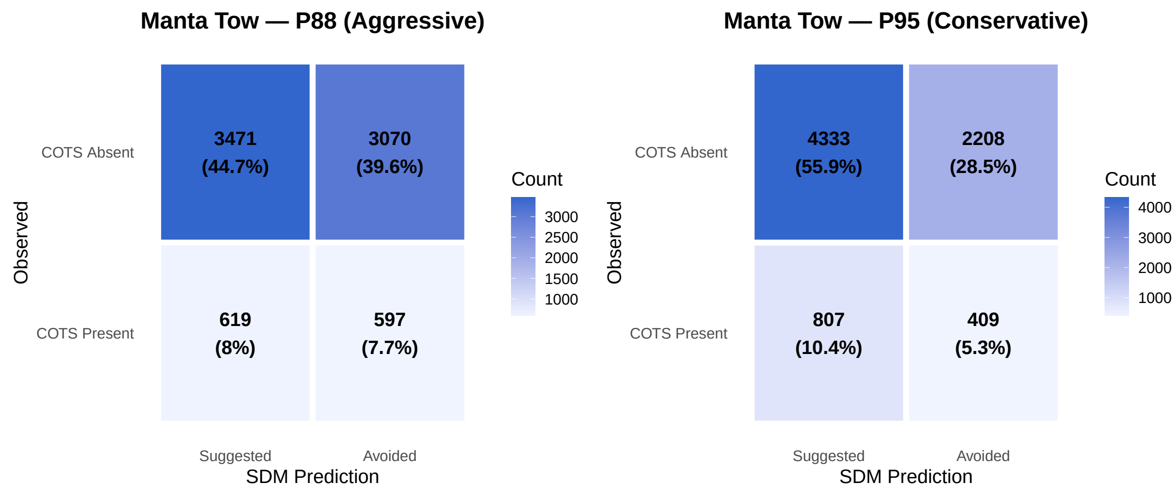
Table 8: Consolidated Scenario Comparison: Manta Tow + Cull Dive (All Top 30 Reefs, 2025)

Scenario	Total_Tows	Tows_Suggested	Tows_Avoided	Hit_Rate	Miss_Rate	Total_Dives	Dives_Avoided	Total_Savings	COTS_Missed	Pct_COTS_Missed
P88 (Aggressive)	7,757	4,090	3,667	0.15	0.16	1,658	532	1,465,470.0[illegible]	5,342	17.56
P95 (Conservative)	7,757	5,140	2,617	0.16	0.16	1,658	321	982,885.8[illegible]	2,911	9.57

Confusion Matrices

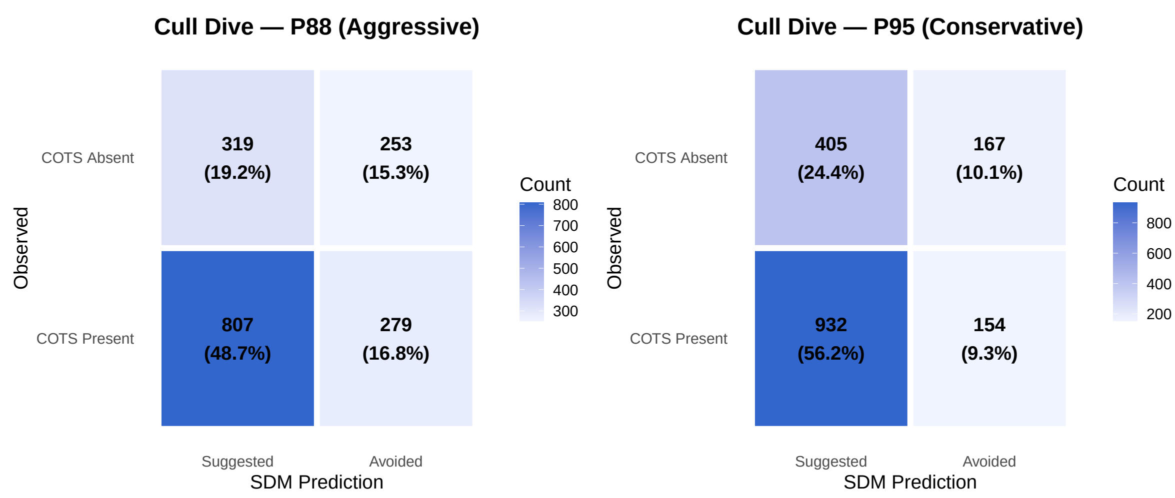
```
# Function to build and plot a confusion matrix
plot_confusion_matrix <- function(predicted_suggested, observed_presence, title) {
  cm_df <- data.frame(
    Predicted = factor(ifelse(predicted_suggested, "Suggested", "Avoided"), levels = c("Sugge
    Observed  = factor(ifelse(observed_presence, "COTS Present", "COTS Absent"), levels = c("
  )
  cm_counts <- cm_df %>%
    count(Predicted, Observed) %>%
    group_by() %>%
    mutate(pct = n / sum(n) * 100)

  ggplot(cm_counts, aes(x = Predicted, y = Observed, fill = n)) +
    geom_tile(color = "white", linewidth = 2) +
    geom_text(aes(label = paste0(n, "\n(", round(pct, 1), "%)") ), size = 5, fontface = "bol
    scale_fill_gradient(low = "#f0f4ff", high = "#3366cc") +
```

```r
    theme_minimal(base_size = 14) +
    labs(title = title, x = "SDM Prediction", y = "Observed", fill = "Count") +
    theme(
      plot.title = element_text(face = "bold", hjust = 0.5),
      panel.grid = element_blank()
    )
}

# Manta Tow Confusion Matrices
p1 <- plot_confusion_matrix(
  manta_eval_2025$sug_Tow_P88, manta_eval_2025$presence_any,
  "Manta Tow - P88 (Aggressive)"
)
p2 <- plot_confusion_matrix(
  manta_eval_2025$sug_Tow_P95, manta_eval_2025$presence_any,
  "Manta Tow - P95 (Conservative)"
)

gridExtra::grid.arrange(p1, p2, ncol = 2)
```




```r
# Cull Dive Confusion Matrices (is_non_problematic as "COTS Absent" proxy)
p3 <- plot_confusion_matrix(
  !cull_eval_2025$avoid_Tow_P88, !cull_eval_2025$is_non_problematic,
  "Cull Dive - P88 (Aggressive)"
)
p4 <- plot_confusion_matrix(
  !cull_eval_2025$avoid_Tow_P95, !cull_eval_2025$is_non_problematic,
```

```
  "Cull Dive - P95 (Conservative)"
)

gridExtra::grid.arrange(p3, p4, ncol = 2)
```



5. Step D: Visual Mapping

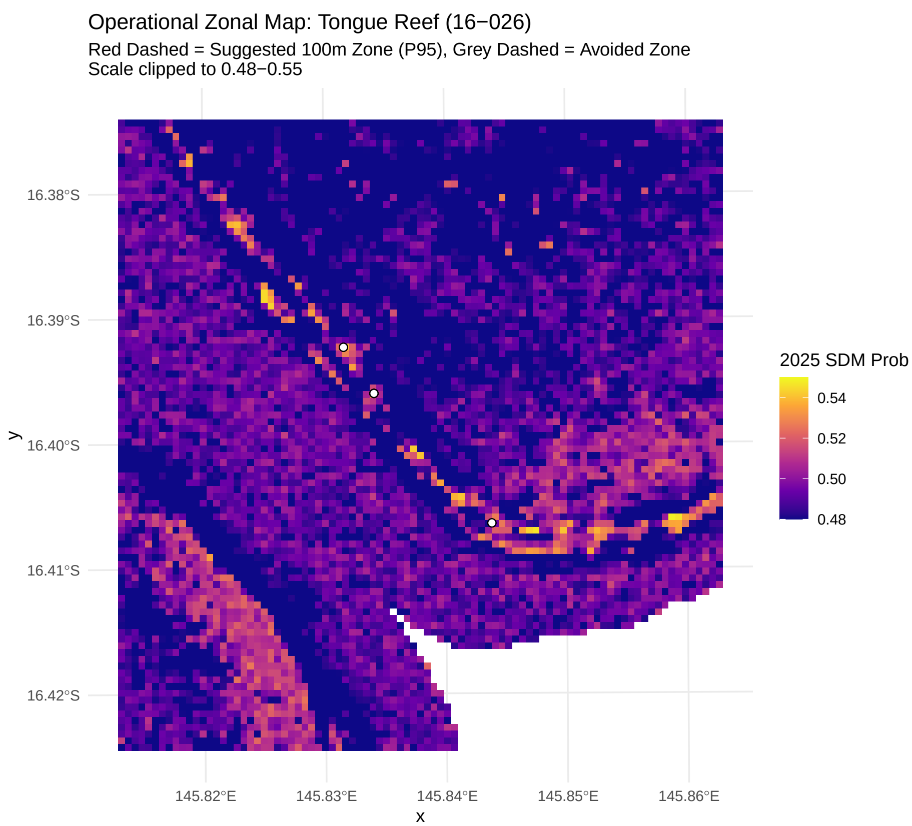
Below are maps for each reef showing the underlying 2025 SDM probability. The operational units displayed are the **100m Buffered Tow Regions**, categorized by the **95th Percentile (P95) Threshold** (Red = Suggested, Grey = Avoided). Individual manta tows and cull dive locations are also plotted to show what falls inside the suggested operational areas.

```
plot_reef_map <- function(r, tow_sf, dive_sf, region_sf, reef_name, p_min, p_max) {
  # Bounding box around features
  bb_t <- if(nrow(tow_sf) > 0) st_bbox(tow_sf) else NULL
  bb_d <- if(nrow(dive_sf) > 0) st_bbox(dive_sf) else NULL

  xmin <- min(c(bb_t["xmin"], bb_d["xmin"]), na.rm=TRUE)
  ymin <- min(c(bb_t["ymin"], bb_d["ymin"]), na.rm=TRUE)
  xmax <- max(c(bb_t["xmax"], bb_d["xmax"]), na.rm=TRUE)
  ymax <- max(c(bb_t["ymax"], bb_d["ymax"]), na.rm=TRUE)

  pad <- 2000 # 2km padding
  ext_map <- terra::ext(xmin - pad, xmax + pad, ymin - pad, ymax + pad)

  rc <- terra::crop(r, ext_map)
```

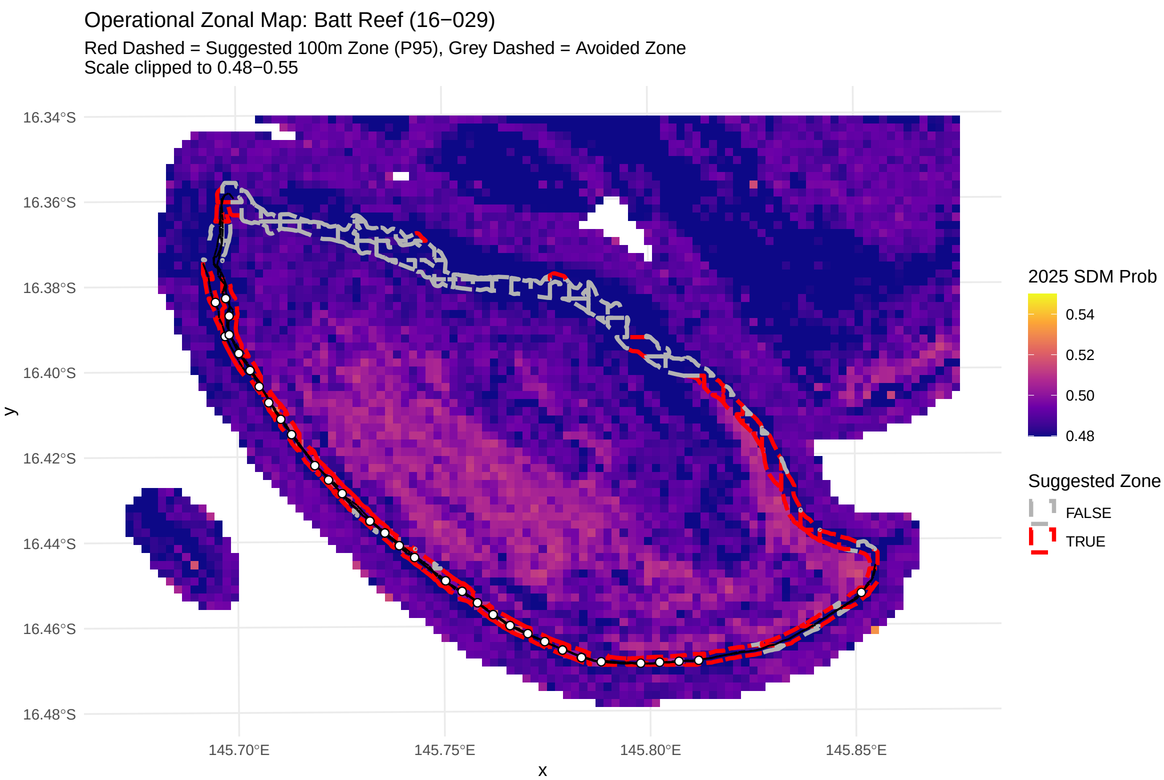
```r
  # Scale aggregate factor to keep cell count manageable (~10,000 pixels) for fast plotting
  fact <- ceiling(sqrt(terra::ncell(rc) / 10000))
  if(fact > 1) {
    rc <- terra::aggregate(rc, fact = fact, fun = "mean", na.rm = TRUE)
  }

  rdf <- as.data.frame(rc, xy = TRUE, na.rm = TRUE)
  names(rdf)[3] <- "prob"

  # Determine if region is suggested based on P95 threshold
  region_sf <- region_sf %>%
    mutate(is_suggested = !is.na(reg_p95) & reg_p95 >= thresh_2025_p95)

  gg <- ggplot() +
    geom_raster(data = rdf, aes(x = x, y = y, fill = prob)) +
    # Plot Operational Zones
    geom_sf(data = region_sf, aes(color = is_suggested), fill = NA, linewidth = 1.2, linetype
    scale_color_manual(values = c("TRUE" = "red", "FALSE" = "grey70")) +
    # Plot Tows and Dives
    geom_sf(data = tow_sf, color = "black", linewidth = 0.5) +
    geom_sf(data = dive_sf, shape = 21, fill = "white", color = "black", size = 2) +
    scale_fill_viridis_c(
      option = "C",
      na.value = NA,
      limits = c(p_min, p_max),
      oob = scales::squish
    ) +
    theme_minimal() +
    labs(
      title = paste("Operational Zonal Map:", reef_name),
      subtitle = paste0("Red Dashed = Suggested 100m Zone (P95), Grey Dashed = Avoided Zone\r
      fill  = "2025 SDM Prob",
      color = "Suggested Zone"
    )
  return(gg)
}

# Generate plots for each reef
cull_sf_2025 <- cull_sf %>% filter(as.Date(SurveyDate) >= as.Date("2025-01-01"))

for(r_name in reef_select) {
  t_sf <- manta_eval_2025 %>% filter(ReefName == r_name)
```

```
  d_sf <- cull_sf_2025 %>% filter(ReefName == r_name)
  r_sf <- tow_regions %>% filter(ReefName == r_name)

  if(nrow(t_sf) > 0 || nrow(d_sf) > 0) {
    message(paste("Processing map for:", r_name))
    cat("\n### ", r_name, "\n")
    p <- plot_reef_map(prob_year2025, t_sf, d_sf, r_sf, r_name, p_min_2025, p_max_2025)
    print(p)
    cat("\n")
  }
}
```

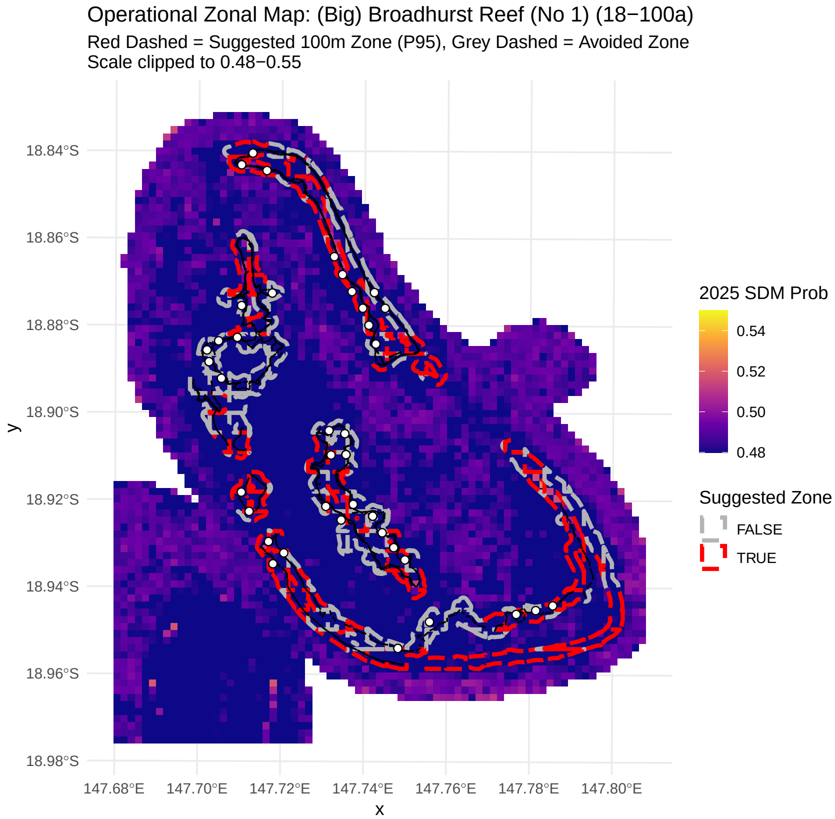
Tongue Reef (16-026)



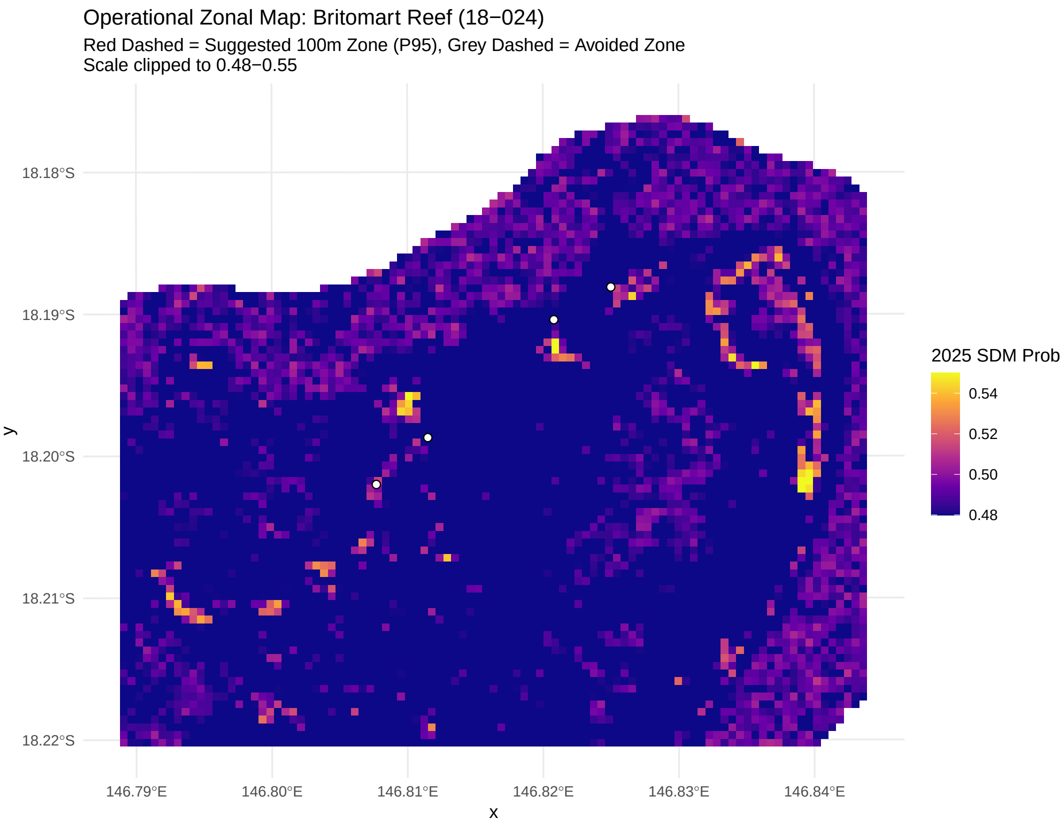
Batt Reef (16-029)



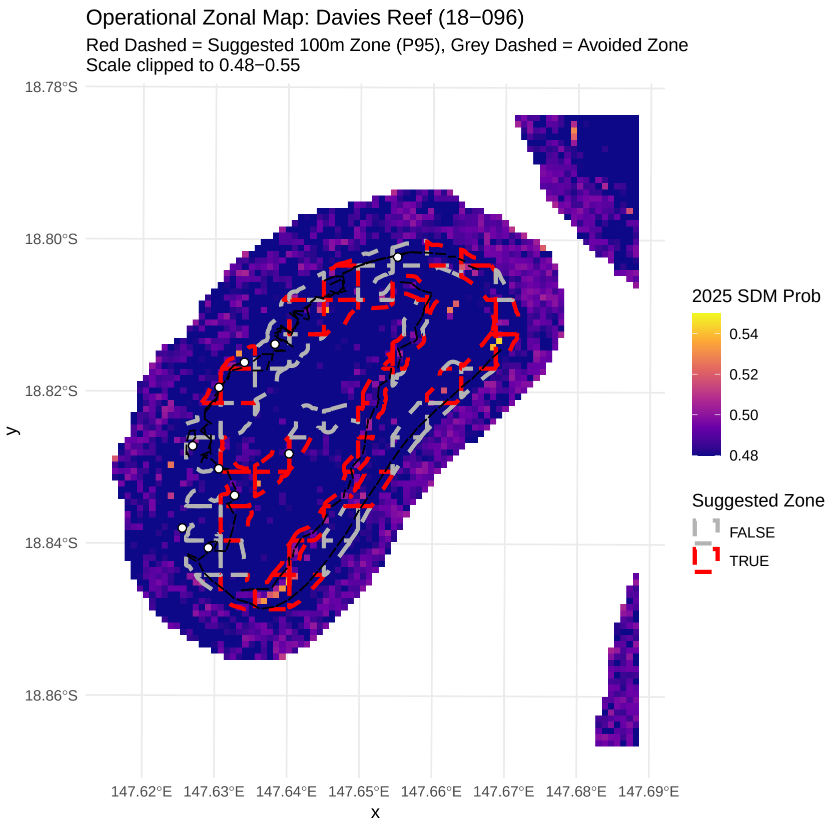
(Big) Broadhurst Reef (No 1) (18-100a)



Britomart Reef (18-024)



Davies Reef (18-096)



Heron Reef (23-052a)